Rita Voice v3

Real iPad Test Failure Analysis and Remediation Plan

Production evidence • latency timeline • root-cause analysis • Claude-ready implementation brief

TEST CONTEXT

Real user · iPad · Safari · ritajet.com · 13 October 2026

CONFIDENTIALITY

No API keys, tokens, personal identifiers, or private user recordings are included.

Prepared as an engineering handoff for Claude / external code review

Executive summary

الملخص التنفيذي بالعربية

هذا التقرير مبني على اختبار حقيقي أجراه المستخدم على جهاز iPad ومن متصفح Safari في موقع RitaJet الإنتاجي، وليس على محاكاة نصية فقط.

الميكروفون وSoniox عملا بصورة صحيحة، وظهرت الجملة المنطوقة كنص. كما ولّد Groq الرد وحوّلت Cartesia الرد إلى صوت PCM غير صامت داخل عامل Modal.

رغم ذلك لم يسمع المستخدم أي صوت. أقوى سبب مثبت في الكود هو أن الواجهة لا تربط مسار الصوت البعيد القادم من Daily بعنصر صوت دائم، ولا تستدعي تشغيله مع معالجة قيود Safari.

في آخر دور أضاف Smart Turn حوالي 1.59 ثانية لأنه صنّف جملة مكتملة على أنها غير مكتملة. بعد اعتماد نهاية الدور، احتاج المسار حتى أول صوت تقني قرابة 0.31 ثانية فقط.

كذلك لا يظهر جواب الشات إلا بعد توقف ريتا عن الكلام، لذلك يبدو النص بطيئاً رغم أنه كان جاهزاً مبكراً. الحلول المقترحة تفصل تشغيل الصوت، نهاية الدور، وعرض النص إلى مراحل قابلة للقياس.

الأولوية: إصلاح تشغيل المسار الصوتي على iPad، ثم تقليل انتظار Smart Turn بحذر، ثم إظهار النص قبل اكتمال تشغيل الصوت، مع اختبارات قبول على الجهاز الحقيقي.

النتيجة: مزودات الصوت تعمل؛ الخلل الأهم حالياً في تشغيل الصوت داخل واجهة iPad، ثم منطق نهاية الدور.

Bottom line
The speech providers are not the primary failure in this session. The worker generated non-silent Cartesia PCM and signalled bot speech, while the iPad played nothing. The highest-confidence fault is the missing browser-side remote-track playback path. A separate Smart Turn false-negative added approximately 1.59 seconds to the final turn.

What this report proves — and what it does not

- It documents a real production test performed by the user on an iPad in Safari, not a text-only or synthetic-provider benchmark.
- It proves that microphone capture, Soniox transcription, Groq generation, and Cartesia PCM generation executed during the session.
- It does not claim p95 reliability, universal Safari behaviour, or accent quality from one session.
- The report separates measured facts, code-supported inferences, and recommended experiments so reviewers can challenge each conclusion.

1. Current production architecture

Rita is a multi-stage streaming voice system. The browser sends microphone audio over Daily WebRTC. A warm worker on Modal receives that audio, Soniox streams transcription, Silero plus Smart Turn decide when the turn is complete, Groq streams the reply, Cartesia produces PCM audio, and Daily returns a remote audio track to the browser.



Component	Current role	Observed status in real test
iPad Safari	Captures the user's microphone; receives the bot's Daily track; should render and play it.	Microphone worked. No audible bot audio.
Daily WebRTC	Bidirectional transport between browser and worker.	Room join and participant audio capture succeeded.
Modal	Runs the Rita worker and keeps the voice stack warm.	Worker started and joined the Daily room.
Soniox EU	Low-latency multilingual streaming STT.	Final transcript was emitted correctly.
Silero + Smart Turn	Detects speech boundaries and decides turn completeness.	One fast COMPLETE decision; one false INCOMPLETE decision.
Groq	Streams Rita's text using openai/gpt-oss-120b.	First model response arrived quickly.
Cartesia	Sonic 3.6 TTS using the selected Layan voice; streams PCM.	First chunk and non-silent PCM were produced.
Lovable UI	Orchestrates the client, messages, metrics and status.	Displayed transcript and reply text; did not audibly play the remote track.

Streaming behaviour

Rita does not wait for the entire LLM response. Groq streams text; once the first speakable clause is available, the worker sends it to Cartesia while Groq may continue generating. However, audio cannot begin before end-of-turn commitment, the first useful clause, and Cartesia's first PCM.

2. Test method and evidence chain

The user initiated a genuine Rita lesson in production from an iPad. The screenshot confirms the live production interface, microphone permission, live transcript, and Rita's visual speaking state. The analysis then correlated the client-visible behaviour with the Lovable server request logs and the Modal worker logs from the same session window.

> Real-user test declaration
> This was not a synthetic audio probe. The user spoke into the iPad microphone and personally verified that no sound was audible. That human observation is the authoritative endpoint for playback; a worker-side PCM event alone cannot prove that a person heard audio.

Evidence source	What it established	Limit
iPad Safari observation	The user spoke, transcript appeared, UI entered a speaking/listening state, but the speaker output remained silent.	Human timing was not frame-accurately recorded.
Lovable worker logs	The production start endpoint returned HTTP 200 and the Modal start call took about 2.760 seconds.	Browser import time before the request is not included.
Modal worker logs	Daily join, Soniox final text, Smart Turn decisions, Groq first response, Cartesia chunks, PCM and speaking events.	A worker event proves generation/transport intent, not iPad playback.
Current client code	No onTrackStarted remote audio attachment, no persistent audio element, no explicit play() handling; chat commit occurs at bot stop.	Code inference must still be confirmed by a patched iPad run.
Daily transport package	Dispatches remote-track lifecycle callbacks and does not itself provide an application audio play() call.	Bundle inspection reflects the installed version.

Confidence vocabulary

- Confirmed: directly present in logs, code, or the user's real-device observation.
- High-confidence inference: code and behaviour agree, but a post-fix device test is required for closure.
- Secondary: relevant operational issue that did not cause this session's principal failure.

3. Startup latency: approximately 7.1 seconds

Startup latency is separate from conversational latency. The observed path included the server's Modal start request, worker setup, Daily join, and client-ready propagation. Some stages overlap and use different clocks, so the values below should not be naively added beyond the measured end-to-end total.

Milestone	Observed time / interval	Interpretation
Production start endpoint	HTTP 200	The previous 503 configuration failure was resolved for this test.
Lovable → Modal start	2,760 ms	Server-side upstream request duration.
Modal session start	20:47:27.179	Worker session began.
Daily transport created	20:47:27.659	Transport setup was underway.
Cartesia WebSocket connected	20:47:28.364	TTS connection was established before conversation.
Bot joined Daily	20:47:29.859	Worker became a room participant.
Participant audio captured	20:47:29.867	Inbound microphone transport was active.
Client ready	20:47:31.116	Client-facing readiness arrived; total about 7.1 s from the start request window.

Measurement gap
RitaRealtimeV3 dynamically imports PipecatClient and DailyTransport after the user presses Start, but sessionStartedAt is assigned after those imports. Therefore the displayed startup metric can under-report the user's true click-to-ready wait.

Likely startup improvements

- Begin the authoritative timer at the first line of the Start-button handler.
- Preload Pipecat and Daily bundles on page idle or when the Rita page becomes visible.
- Record distinct browser milestones: click, imports ready, start response, transport connected, remote track started, audio play resolved.
- Only after measurement, consider parallelising room credential creation and worker preparation. Do not change providers to solve a UI measurement gap.

4. Conversational latency: the final turn

For the test utterance “Hey, how are you?”, the pipeline did not stall in Groq or Cartesia. The dominant delay happened before the turn was committed. Smart Turn labelled the utterance INCOMPLETE, causing the 1.5-second fallback timer to fire.

Stage	Elapsed from inferred speech stop	Increment / evidence
User speech stops	0 ms	Inferred VAD stop around 20:47:50.461.
Soniox final transcript	+34 ms	Final text emitted around 20:47:50.495; Soniox service TTFB recorded about 233 ms.
Smart Turn decision	+77 ms	Decision was INCOMPLETE — inconsistent with the clearly complete test question.
Fallback turn commit	+1,589 ms	user_turn_stop_timeout=1.5 caused delayed commitment.
Groq first response	+1,690 ms	About 101 ms after committed speech_end.
First useful answer / clause	+1,812 ms	About 222–224 ms after commitment.
Cartesia first text sent	+1,814 ms	Clause entered TTS immediately.
Cartesia first non-silent PCM	+1,898 ms	About 85 ms Cartesia TTFB; leading silence was trimmed.
Bot speaking signal	+1,900 ms	Technical first outgoing audio signal, not proof of audible iPad sound.

Delay group	Time	Share of ~1.90 s
End-of-turn waiting	~1.589 s	~83.7%
Post-commit generation to first outgoing audio	~0.311 s	~16.3%

Important comparison
An earlier Arabic turn was much faster: Smart Turn returned COMPLETE, the turn committed roughly 151 ms after VAD stop, Groq took about 120 ms, Cartesia about 89 ms, and the bot speaking signal occurred around 398 ms after commitment. The system is capable of a fast path; the final turn exposed a variable end-of-turn failure.

5. Root cause #1 — no audible audio on iPad

Severity: Critical · Confidence: High
Backend audio generation succeeded, but the browser-side client has no explicit remote audio track playback path. This is the first issue Claude should fix.

Evidence supporting this conclusion

- Cartesia connected, received the first clause, produced a chunk, and produced non-silent PCM.
- The worker emitted a bot-speaking signal, so the TTS path was not failing silently at the provider.
- The user heard nothing on the actual iPad even though the UI and transcript continued to update.
- The current RitaRealtimeV3 client creates PipecatClient and DailyTransport but does not register onTrackStarted for remote bot audio.
- There is no persistent HTML audio element, no MediaStream attachment, no explicit audio.play(), and no handling of a rejected play promise.
- The installed Daily transport dispatches track callbacks; it does not replace the application's responsibility to render/play the track.

Required client behaviour

The application should keep a persistent audio element and attach the bot's remote audio track when it starts. Playback must be attempted, the returned Promise must be handled, and Safari autoplay rejection must produce an explicit “Tap to enable sound” control rather than silent failure.

Conceptual implementation (adapt to the installed Pipecat callback types):

```
const remoteAudio = audioRef.current; onTrackStarted: async (track, participant) => { if (track.kind !== 'audio' || participant?.local) return; remoteAudio.srcObject = new MediaStream([track]); try { await remoteAudio.play(); metric('audio_playing'); } catch (error) { metric('audio_play_blocked'); showEnableSound(); } }
```

The audio element should use autoplay and playsInline, remain mounted for the session, and be reused across track restarts. A track-ended handler should clear only the matching stream, not destroy the entire audio context.

6. Root cause #2 — Smart Turn false INCOMPLETE

Severity: High · Confidence: Confirmed for the measured turn
The clearly complete English question was classified as incomplete. The configured user_turn_stop_timeout=1.5 then added approximately 1.59 seconds before any LLM/TTS work could begin.

Recommended policy

- Keep Smart Turn; do not replace it with a fixed aggressive timeout globally.
- Add a bounded fallback for high-confidence final transcripts that look syntactically complete: approximately 450–650 ms, tuned by language and punctuation/confidence signals.
- Retain the longer 1.5-second path for genuine hesitation, conjunction endings, unfinished clauses, and noisy/low-confidence transcripts.
- Log the reason for each commit: smart_complete, bounded_complete_fallback, or long_incomplete_fallback.
- Build multilingual tests for Jordanian Arabic, German, English, code-switching, filler sounds, pauses, and user interruption.

7. Root cause #3 — chat waits for audio completion

Severity: Medium–High · Confidence: Confirmed in client code
The UI calls commitHeardReply from onBotStoppedSpeaking. In the measured final turn, the text was available much earlier, but the visible assistant message could be delayed until roughly seven seconds after the user's speech stopped because it waited for the utterance to finish playing.

Recommended separation of concerns

- Show or stream the assistant text when the LLM/TTS text becomes authoritative; do not wait for playback completion.
- Track separate states: generated text, queued speech, actually played/heard speech, and interrupted speech.
- For learning extraction, only mark the spoken prefix as heard when playback progress confirms it; do not pretend the interrupted suffix was heard.
- Keep onBotStoppedSpeaking for final playback accounting, not for first message visibility.

8. Secondary findings

Finding	Impact	Recommended action
Startup metric begins after dynamic imports	Understates true click-to-ready latency and hides bundle loading regressions.	Start timer immediately; add imports_ready milestone; preload bundles.
iOS startup delay in transport path	Contributes to readiness after join; not the cause of silent playback.	Measure separately before changing; avoid removing compatibility delay blindly.
Missing SUPABASE_SERVICE_ROLE_KEY warnings	Can weaken persistence/admin diagnostics. Session and metrics endpoints still returned 200 in this test.	Fix Lovable server secret exposure separately; do not mislabel it as the no-audio root cause.
Single real session	Excellent for reproducing the critical failure, insufficient for p95 or regression confidence.	After the fix, run a small deterministic device matrix, then a larger latency sample.
No frame-accurate external recording	Worker timestamps cannot prove the exact human audible onset.	For final validation, screen-record and preferably film the iPad from a second device.

What should not be changed in this repair

- Do not replace Soniox, Groq, Cartesia, Daily, or Modal based on this session.
- Do not add a Gemini fallback; Cartesia generated PCM successfully.
- Do not shorten every turn to 200 ms; that would cut hesitant and multilingual speakers.
- Do not use worker PCM timestamps as proof that Safari played audio.
- Do not couple the audio fix to database or admin-secret refactoring.

Operational interpretation
This is a recoverable integration problem, not evidence that real-time voice is impossible. The measured providers were fast after turn commitment; the main defects are browser playback wiring and end-of-turn policy.

9. Claude implementation plan

Claude should implement the following in separate, reviewable commits so a regression can be isolated and reverted.

Instrument before changing timing
Add true click-to-ready metrics and browser audio milestones: remote_track_started, audio_element_attached, audio_play_called, audio_play_resolved, audio_play_blocked, audio_playing, audio_stalled, audio_ended. Never include tokens or transcripts in these events.

Fix remote audio playback
Add a persistent audio element, register onTrackStarted, attach the bot MediaStreamTrack, call play(), catch NotAllowedError, and present a user-gesture recovery control. Perform an audio unlock during the Start click before asynchronous imports/network work.

Validate on iPad before latency tuning
Run the same real-device test. Require audible sound, audio_play_resolved/playing events, and no provider changes. If it still fails, inspect the received track state, enabled/muted flags, output volume, and Safari media diagnostics.

Decouple text visibility from playback completion
Display the authoritative assistant text as it streams or when the assistant message closes. Preserve a separate spoken/heard cursor for interruption-safe learning logic.

Introduce bounded end-of-turn fallback
When Smart Turn says INCOMPLETE but the final transcript is high-confidence and linguistically complete, use a 450–650 ms cap. Preserve 1.5 s for genuinely incomplete turns. Add reason-coded metrics and multilingual unit/integration tests.

Optimise startup after honest measurement
Preload client bundles, retain safe iOS behaviour, then evaluate room/worker parallelism using the new milestones. Keep the existing providers during this phase.

Likely files / components

Path	Expected change
src/components/rita/RitaRealtimeV3.tsx	Remote-track playback, audio unlock, UI recovery, metrics, early text visibility, true startup timer.
voice-agent/bot.py	Bounded end-of-turn policy and reason-coded timing; keep current providers.
src/lib/rita-v3-heard.ts	Separate generated, played, and heard text semantics without crediting interrupted audio.
Tests around Rita client and worker	Mock remote tracks/play rejection; multilingual turn-boundary cases; interruption and reconnect coverage.

10. Verification and acceptance criteria

The repair is complete only when the observable user experience and the internal evidence agree.

Area	Acceptance check	Pass condition
Audibility	Real iPad Safari session with the same Start flow.	The user hears Rita on every tested turn; remote track and play-resolved events exist.
Autoplay recovery	Force or simulate play() rejection.	A clear Tap to enable sound control appears; tapping it resumes playback without restarting the lesson.
Normal-turn latency	Arabic, German and English complete questions.	User speech stop to first audible Rita sound is targeted below 1.0 s on a warm healthy path.
Fallback latency	Question that Smart Turn misclassifies as incomplete.	Bounded complete fallback targets below 1.2 s without clipping real incomplete speech.
Chat visibility	Observe assistant text while audio is still playing.	Text appears promptly (target: within 500 ms of authoritative first LLM text), not only at bot stop.
Interruption	Interrupt Rita mid-utterance.	Playback stops promptly; only the played/heard prefix is credited.
Startup	Measure from the first Start-button instruction.	Every phase is present; warm click-to-ready target is initially below 5 s, then tuned from evidence.
Regression	Run provider and language matrix.	No loss of Soniox transcript, Groq reply, Cartesia PCM, language lock, Flashcard gate, or Daily reconnect.

Minimum test matrix

- Jordanian Arabic complete question, incomplete phrase, hesitation, and brief filler sound.
- German complete and incomplete questions; explicit language switch.
- English question matching the failed test utterance.
- Arabic sentence containing one German term without accidental language switching.
- Remote track restart, network reconnect, output-device mute/unmute, and Safari play rejection.
- User interruption before and after first audio; verify spoken-prefix accounting.

Release gate
Do not declare success from Modal logs alone. A real iPad must produce audible sound, and the reportable first-audio metric must distinguish worker PCM, browser track arrival, play() success, and human-audible confirmation.

11. Reviewer brief for Claude

Use the following brief together with the repository. It is intentionally narrow so the reviewer does not redesign the whole platform before fixing the reproduced failure.

Primary request
Review the current Rita Voice v3 client and worker. First, implement and test explicit remote Daily audio playback on iPad Safari, including autoplay rejection recovery and stage-level metrics. Second, decouple assistant text visibility from playback completion. Third, design a bounded multilingual Smart Turn fallback that reduces the measured 1.59-second false-INCOMPLETE wait without cutting genuinely unfinished speech. Preserve Soniox, Groq, Cartesia, Daily, Modal, the selected Layan voice, language-lock behaviour, interruption, and Flashcard gating.

Questions Claude must answer in its review

- Does the installed Pipecat/Daily client version require application-owned audio attachment exactly as inferred? Cite the local types/source.
- What is the safest iPad Safari audio-unlock sequence that remains inside the Start-button user gesture?
- How should remote track replacement/reconnect be handled without duplicate playback?
- Which event provides authoritative assistant text early enough for UI display, and how should interrupted speech be represented?
- How can transcript confidence, punctuation and language-specific completion cues bound a false-INCOMPLETE timeout without harming slow speakers?
- Which new automated tests prove the fixes without relying only on worker logs?

Repository reference points

src/components/rita/RitaRealtimeV3.tsx — client creation around line 424; DailyTransport around 458; onBotStoppedSpeaking/commitHeardReply around 656–659.

voice-agent/bot.py — LocalSmartTurnAnalyzerV3 import; VAD stop_secs=0.2 around line 234; user_turn_stop_timeout=1.5 around line 246; Cartesia service around line 217.

src/lib/rita-v3-heard.ts — semantics for what was generated, spoken, interrupted and credited as heard.

12. Sources, caveats and final diagnosis

External references

Reference	Relevance
Pipecat p2p WebRTC client example	Shows application handling of onTrackStarted and attachment of bot audio to an HTML audio element. https://github.com/pipecat-ai/pipecat-examples/blob/main/p2p-webrtc/pipecat-cloud/client/src/app.ts
Pipecat Daily transport README	Documents the Daily web transport used by the browser client. https://github.com/pipecat-ai/pipecat-client-web-transports/blob/main/transports/daily/README.md
WebKit — A closer look into WebRTC	Explains Safari media/autoplay behaviour for MediaStream playback and capture. https://webkit.org/blog/7763/a-closer-look-into-webrtc/
Apple — Delivering video content for Safari	Safari media delivery and playback considerations. https://developer.apple.com/documentation/webkit/delivering-video-content-for-safari
MDN — Autoplay guide	play() promise rejection and user-gesture recovery patterns. https://developer.mozilla.org/en-US/docs/Web/Media/Guides/Autoplay
Pipecat Smart Turn overview/source	Turn completeness concepts and implementation context. https://github.com/pipecat-ai/docs/blob/main/api-reference/server/utilities/turn-detection/smart-turn-overview.mdx

Caveats

- Timestamps came from multiple sources; only intervals measured inside one clock domain are treated as precise.
- The startup total is approximate because the browser did not record every pre-request phase.
- The audio-playback cause is high-confidence, not closed, until the patched client succeeds on the same iPad.
- One session can identify reproduced failures and first bottlenecks; it cannot establish p50/p95 production statistics.

Final diagnosis

Rita's providers completed the measured response path quickly after turn commitment. The real user heard no sound because the client likely never rendered/played the remote Daily audio track. Perceived slowness was then amplified by a Smart Turn false-INCOMPLETE decision and by a UI policy that waits for bot playback to finish before committing the chat message. Fix those three issues in that order, verify on the real iPad, and only then optimise startup or reconsider providers.

Document status: Engineering diagnosis for external review. No credentials, tokens, session identifiers, private recordings, or user account data included.